MONOCHROMATIC DISTANCES IN FINITE COLORINGS OF THE PLANE

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Abstract. We show that every coloring of the plane with finitely many colors has, for all sufficiently large n, two points of the same color at distance 2^n, and two at distance $n!$, answering questions of Soifer and of Kahle. More generally no finite coloring avoids an unbounded set of algebraic distances. We also show that the number of colors needed to avoid r distances in the plane grows exponentially in r, answering Erdős Problem #706 in the negative, with explicit distance sets: every coloring avoiding the distances $1, q, \dots, q^{K-1}$ uses more than $\frac{22}{15}(2.5^K - 1)$ colors when $q \geq 4096$ is algebraic. These results, and a quantitative form of Soifer's question, are formally verified in Lean 4 with only the standard axioms.

The proofs combine the transfer principle of OpenAI's proof that the plane is not five-colorable, extended to sets of algebraic distances, with explicit Fourier certificates. They also give quantitative bounds: every k-coloring of the plane has a monochromatic odd distance below $18k^2$, a monochromatic distance $2^i < k^{6.2}$, and a monochromatic distance in any set meeting every interval $[X, 2X]$ below a fixed power of k; in $\mathbb{R}^3$ the odd distance can be taken at most k. For two distances, $\chi(\mathbb{R}^2, \{1, d\}) \geq 7$ for every algebraic $d \geq 7$, and ≥ 10 for $d \geq 150$.

1. Introduction

For a set $D \subset (0, \infty)$ let $\chi_D(\mathbb{R}^n) = \chi(\mathbb{R}^n, D)$ be the least number of colors in a coloring of $\mathbb{R}^n$ with no two points of the same color at a distance in D; no regularity is imposed on the color classes. For $D = \{1\}$ this is the chromatic number of $\mathbb{R}^n$; in the plane OpenAI [17] recently proved $\chi(\mathbb{R}^2) \geq 6$. Its first step is a transfer theorem: a proper coloring of the plane yields a measurable coloring whose monochromatic unit pairs form a null set. In a companion paper [5] we extend the transfer to every dimension and to every countable set of algebraic distances, with a Lean formalization; Crocker [4] independently stated the same extension conditionally on [17, Theorem 2.3]. This paper develops the consequences for colorings that avoid several distances.

Three questions motivate it. Erdős asked whether $\sup_{|D|=r} \chi_D(\mathbb{R}^2)$ grows polynomially in r [14, 10]; the best lower bound was of order $r\sqrt{\log r}$ [6, 16]. Davies proved that every finite coloring of the plane has a monochromatic pair at an odd distance [7], and Davies, McCarty and Pilipczuk did the same for primes and polynomial values [8]. They asked whether powers of 2 (Soifer) and factorials (Kahle) behave likewise, and remarked that they knew of no infinite subset of $\mathbb{N}$ that a finite coloring of the plane can avoid. Davies [9] answered the analogous questions in dimensions $n \geq 4$ for distances with rational squares.

Relation to [4]. Crocker's note states the transfer for algebraic distance sets, the unavoidability of unbounded sets, and the exponential growth for widely separated distances, conditionally on [17, Theorem 2.3] and without explicit distances or quantitative bounds. Here the qualitative statements are machine-checked, which makes them unconditional relative to the Lean kernel since [17, Theorem 2.3] is itself formalized, and the distance sets and bounds are explicit.

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2. Transfer for sets of algebraic distances

For a finite set $D \subset (0,\infty)$ let $\chi_D(\mathbb{R}^n) = \chi(\mathbb{R}^n, D)$ be the least number of colors in a coloring of $\mathbb{R}^n$ with no monochromatic pair at a distance in D, and let $m_D(\mathbb{R}^n)$ be the supremal upper density of a measurable set with no two points at a distance in D. Let $F_n(k) = \sup_{|D|=k} \chi_D(\mathbb{R}^n)$. Erdős [14] asked whether $F_2(k)$ grows polynomially in k (Bukh [2] records this as a conjecture), and Croft, Falconer and Guy [6, Problem G11] asked whether it grows exponentially. In the plane the best lower bound was $F_2(k) \geq ck\sqrt{\log k}$, from the square grid [6, 16]. Bukh [2] answered the measurable version: $m_D(\mathbb{R}^n) \geq m_1(\mathbb{R}^n)^k$ for every k-element D, and $m_{t_1 D_1 \cup \cdots \cup t_k D_k}(\mathbb{R}^n) \to \prod_i m_{D_i}(\mathbb{R}^n)$ when all ratios t_i/t_{i+1} tend to infinity. Davies [9, Theorems 1.2, 1.3] transferred this to arbitrary colorings in dimensions $n \geq 4$ for distances with rational squares. Crocker [4, Corollaries 8.2, 8.3] states the following theorem conditionally on [17, Theorem 2.3], including the application to the questions of Soifer and Kahle.

Theorem 2.1. *Let $n \geq 2$.*

1. *For every finite set D of positive real algebraic numbers, $\chi_D(\mathbb{R}^n) \geq 1/m_D(\mathbb{R}^n)$.*
2. *For every $k \geq 1$ and $\varepsilon > 0$ there is R such that $\chi_D(\mathbb{R}^n) \geq (m_1(\mathbb{R}^n) + \varepsilon)^{-k}$ for every set $D = \{d_1 < \cdots < d_k\}$ of positive real algebraic numbers with $d_{i+1}/d_i \geq R$. In particular*
$$F_2(k) \;\geq\; m_1(\mathbb{R}^2)^{-k} \;>\; 4.0486^k \qquad \text{and} \qquad F_3(k) \;\geq\; m_1(\mathbb{R}^3)^{-k} \;>\; 6.523^k.$$
3. *For every coloring of $\mathbb{R}^n$ with finitely many colors there is $R > 0$ such that every positive real algebraic $r \notin [1/R, R]$ is the distance between two points of the same color (which color may depend on r).*

Proof. (1) This is the transfer for sets of algebraic distances of [5], formalized there as `euclidean_transfer_set`; we recall the argument. One runs the transfer proof of [5] (and, in the plane, of [17, Section 4]) with the unit sphere replaced by the spheres of radii $d \in D$. The one-label transfer of [5] holds with the hypothesis and conclusion stated for $|a| \in D$ and $|x| \in D$, with the same proof. Properness gives $\langle f_i, T_a f_i \rangle = 0$ for algebraic a with $|a| \in D$. The block step uses only the twisted rigidity lemma of [5], whose conclusion holds at every generic a regardless of its length, because [17, Theorem 2.3] gives zero correlation at every nonzero $z \in E$. In odd dimension the axis step uses the curves $t \mapsto (\sqrt{d^2 - t^2}\, u, t)$, $|t| < d$, which consist of algebraic points because d is algebraic and F is real closed, with test functions $\varphi \in C_c((-d, d))$. Algebraic points are dense on each sphere of algebraic radius. The density-point lemma [17, Lemma 4.3] applies to each $d \in D$. So a proper k-coloring yields k measurable D-avoiding sets covering $\mathbb{R}^n$ up to a null set, and one of them has upper density at least $1/k$.

(2) Since m_D is scale-invariant, it depends only on the consecutive ratios of D. If for some ε no R worked, there would be sets D_j with all consecutive ratios at least j and $m_{D_j} > (m_1 + \varepsilon)^k$, contradicting [2, Theorem 1] with $D_i = \{1\}$ (whose distances are listed in decreasing order there). Hence there is R with $m_D(\mathbb{R}^n) \leq (m_1(\mathbb{R}^n) + \varepsilon)^k$ whenever all consecutive ratios are at least R. Choosing the d_i rational shows that $F_n(k) \geq (m_1 + \varepsilon)^{-k}$ for every ε. The numerical bounds use $m_1(\mathbb{R}^2) \leq 0.246998$ [1, Section 8] and $m_1(\mathbb{R}^3) \leq 0.1532996$ [3].

(3) Let the coloring use c colors, and let B be the set of real algebraic $r > 0$ that are not monochromatic distances. Choose k with $(m_1(\mathbb{R}^n) + \varepsilon)^{-k} > c$ for some $\varepsilon > 0$, and let R_k be as in (2). If B were unbounded, it would contain $d_1 < \cdots < d_k$ with $d_{i+1}/d_i \geq R_k$. The coloring avoids $D = \{d_1, \ldots, d_k\}$, so $c \geq \chi_D(\mathbb{R}^n) > c$, a contradiction. If B contained elements arbitrarily close to 0, choose instead $d_k < \cdots < d_1$ in B with $d_i/d_{i+1} \geq R_k$; the same contradiction follows. □

By the de Bruijn–Erdős theorem, $F_2(k)$ is also the supremum of the chromatic numbers of finite graphs in the plane whose edges are the pairs at distances in a k-element set. So (2)

answers Erdős Problem #706 of [10], "is it true that $L(r) \le r^{O(1)}$?", in the negative. Davies [9, Theorem 1.2] states Bukh's limit in the same uniform form for $n \ge 4$.

Part (3) is the analogue for arbitrary colorings of the theorem of Furstenberg, Katznelson and Weiss [15] that a measurable set of positive upper density realizes every sufficiently large distance. In the plane it contains, as special cases, that every finite coloring has a monochromatic pair at an odd distance [7] and at a prime distance [8]. It strengthens these to *all* sufficiently large integers. It also answers the questions of Soifer and of Kahle recorded in [8, Problems 5.4, 5.5]: every finite coloring of the plane has a monochromatic pair at distance 2^j, and one at distance $j!$, for all sufficiently large j. The authors of [8] remark that they do not know a single infinite subset of $\mathbb{N}$ that a finite coloring of the plane can avoid. By (3) there is none, and the same holds for infinite sets of algebraic numbers that are unbounded or accumulate at 0.

3. Fourier certificates and quantitative bounds

Theorem 2.1(1) also turns quantitative density bounds into bounds for arbitrary colorings. Let $D \subset (0,\infty)$ be finite. If real numbers v_d ($d \in D$) satisfy

$$h(t) \;=\; 1 + \sum_{d\in D} v_d\, J_0(dt) \;\ge\; 0 \qquad (t \ge 0), \tag{1}$$

then $m_D(\mathbb{R}^2) \le 1/(1+\sum_d v_d)$ [18]. Since D is bounded, m_D is the supremum of the densities of periodic D-avoiding sets (truncate to a large cube, remove a boundary layer of width $\max D$, and repeat periodically, as in [4, Lemma 2.2]). The radial autocorrelation f of a periodic D-avoiding set of density δ is $f(r) = \int J_0(tr)\, d\mu(t)$ for a positive measure μ of mass δ with $\mu(\{0\}) \ge \delta^2$, and $f(d) = 0$ for $d \in D$. Hence $\delta = \int h\, d\mu \ge h(0)\delta^2$.

Theorem 3.1. *For every $N \ge 1$, $\chi(\mathbb{R}^2, \{1,3,5,\dots,2N-1\}) \ge 1 + N/(2.5\sqrt{N}+1)$. Consequently every coloring of the plane with k colors has two points of the same color at an odd distance less than $18k^2$.*

Proof. Let $F_N(x) = \sum_{|j|<N}(1-|j|/N)e^{ijx} = \sin^2(Nx/2)/(N\sin^2(x/2))$ be the Fejér kernel, and $P(u) = \cos u\, F_N(2u)$. Expanding the product shows $P(u) = \sum_{j=0}^{N-1} c_j \cos((2j+1)u)$ with $c_j \ge 0$ and $\sum_j c_j = P(0) = N$. By $J_0(x) = \frac{1}{\pi}\int_0^\pi \cos(x\sin\theta)\, d\theta$,

$$G(t) := \sum_j c_j J_0((2j+1)t) = \frac{2}{\pi}\int_0^{\pi/2} P(t\sin\theta)\, d\theta \;\ge\; -\frac{2}{\pi}\int_{\{u\in[0,t]:\cos u<0\}} \frac{F_N(2u)\, du}{\sqrt{t^2-u^2}}.$$

The integral is empty for $t \le \pi/2$. Use $F_N(2u) \le \min(N, \pi^2/(4Ns^2))$ with $s = \operatorname{dist}(u, \pi\mathbb{Z})$, $\int F_N(2u)\, du = \pi$ over each period, and $(t^2-u^2)^{-1/2} \le (t(t-u))^{-1/2}$. The peaks of $F_N(2u)$ in $\{\cos u < 0\}$ sit at odd multiples of π, which are 2π apart. A peak u_ℓ with $t - u_\ell \ge \pi$ contributes at most $\pi\sqrt{2}/\sqrt{t(t-u_\ell)}$, and these values of $t-u_\ell$ are at least π and 2π apart, so together they contribute at most $\sqrt{2\pi/t}\,\sum_{m\le t/(2\pi)}(2m-1)^{-1/2} \le \sqrt{2}$. At most one peak lies in $(t-\pi, t+\pi/2)$, since the peaks are 2π apart. Its contribution is bounded by the Hardy–Littlewood rearrangement inequality. The one-sided kernel $y^{-1/2}\mathbf{1}_{y>0}$ has symmetric decreasing rearrangement $(2|y|)^{-1/2}$, so the bound is $t^{-1/2}\int_{\mathbb{R}} \min(N, \pi^2/(4Ns^2))(2|s|)^{-1/2} ds = \frac{8}{3}\sqrt{\pi}\sqrt{N}\, t^{-1/2}$, with $t \ge \pi/2$. Altogether $G(t) \ge -\frac{2}{\pi}\big(\frac{8}{3}\sqrt{\pi}\sqrt{2/\pi}\,\sqrt{N} + \sqrt{2}\big) \ge -(2.5\sqrt{N}+1)$, so (1) holds with $v_{2j+1} = c_j/(2.5\sqrt{N}+1)$. This gives $m_D(\mathbb{R}^2) \le (1 + N/(2.5\sqrt{N}+1))^{-1}$, and Theorem 2.1(1) gives the first claim. For the second, take $N = 9k^2$, for which $N/(2.5\sqrt{N}+1) \ge k$. □

For measurable colorings the order $\sqrt{N}$ is implicit in Steinhardt's spectral argument [11]. For arbitrary colorings, the proof in [7] that the odd-distance graph has infinite chromatic number is not quantitative: it gives no explicit bound on the odd distance.

Theorem 3.2. *Let $q \geq 4096$ be a real algebraic number and $K \geq 1$. Then*

$$\chi\big(\mathbb{R}^2, \{1, q, q^2, \dots, q^{K-1}\}\big) \;\geq\; 1 + \tfrac{22}{15}\big(2.5^K - 1\big).$$

Consequently $\sup_{|D|=k} \chi_D(\mathbb{R}^2) \geq 2.5^k$ with explicit distance sets D, and every coloring of the plane with $k \geq 2$ colors has two points of the same color at a distance $2^i < k^{6.2}$.

Proof. Put $\lambda = 5/2$, $v_j = \frac{11}{5}\lambda^j$, $A = \frac{1}{2}$, $\rho = 0.4028 \geq -\min J_0$, and $\kappa = \sqrt{2/(\pi A)}\,\frac{\lambda/\sqrt{q}}{1-\lambda/\sqrt{q}} \leq 0.046$. We use $J_0 \geq -\rho$, $J_0(x) \geq 1 - x^2/4$, and $|J_0(x)| \leq \sqrt{2/(\pi x)}$ for $x > 0$. Fix $t > 0$ and let m be the largest index $j \leq K-1$ with $q^j t \leq A$, or $m = -1$ if there is none. The terms with $j \leq m$ have arguments at most Aq^{j-m} and contribute at least $\sum_{j\leq m} v_j - \frac{A^2}{4} v_m(1 + 10^{-6})$. The term $j = m+1$ contributes at least $-\rho v_{m+1}$. The terms $j \geq m+2$ have arguments greater than Aq^{j-m-1} and contribute at least $-\kappa v_{m+1}$. Hence

$$h(t) = 1 + \sum_j v_j J_0(q^j t) \;\geq\; 1 + v_m\Big(\tfrac{\lambda(1-\lambda^{-m-1})}{\lambda-1} - \tfrac{A^2}{4}(1+10^{-6}) - \rho\lambda - \kappa\lambda\Big)$$

for $m \geq 0$, and $h(t) \geq 1 - v_0(\rho + \kappa)$ for $m = -1$. The bracket is at least $1.4 - 0.063 - 1.007 - 0.115 > 0$ for $m \geq 1$. For $m = 0$ it is at least $1 - 0.063 - 1.007 - 0.115 > -0.19$, and $1 - \frac{11}{5}\cdot 0.19 > 0$. For $m = -1$, $1 - \frac{11}{5}(0.4028 + 0.046) > 0$. So (1) holds, $\sum_j v_j = \frac{22}{15}(2.5^K - 1)$, and Theorem 2.1(1) applies. For the last claim, rerun the argument with $q = 32$, $\lambda = 1.75$, $v_j = \frac{5}{4}\lambda^j$ and $A = 0.83$. Then $\kappa \leq 0.3923$, and the loss term is $\frac{A^2}{4}(1 - \frac{1}{\lambda q^2})^{-1} \leq 0.1724$. The three cases give brackets $1.5714 - 0.1724 - 0.7049 - 0.6866 > 0$, then $1 + \frac{5}{4}(1 - 0.1724 - 0.7049 - 0.6866) > 0$, then $1 - \frac{5}{4}(0.40276 + 0.3923) > 0$. So $\chi(\mathbb{R}^2, \{32^j : j < K\}) \geq 1 + \frac{5}{3}(1.75^K - 1)$. With $K = \lceil \log_{1.75} k \rceil$ this exceeds k, so some 32^j with $j \leq K-1$ is a monochromatic distance, and $32^{K-1} < k^{5/\log_2 1.75} < k^{6.2}$. □

In $\mathbb{R}^3$ the radial kernel is $\Omega_3(x) = \sin x / x$, and the odd-distance bound becomes linear.

Theorem 3.3. *For every $N \geq 1$, $\chi(\mathbb{R}^3, \{1, 3, \dots, 2N-1\}) \geq 1 + 2N$. Consequently every coloring of $\mathbb{R}^3$ with k colors has two points of the same color at an odd distance at most k.*

Proof. With c_j as in the proof of Theorem 3.1 and $P(u) = \cos u\, F_N(2u)$,

$$G_3(t) := \sum_j c_j\, \Omega_3((2j+1)t) = \frac{1}{t}\int_0^t P(u)\, du = \frac{1}{t}\sum_j c_j \frac{\sin((2j+1)t)}{2j+1} = \frac{\pi}{2t}\,\sigma(t),$$

where σ is the Fejér mean of order $2N$ of the square wave $\operatorname{sgn}(\sin t)$, since $c_j = 2(1 - \frac{2j+1}{2N})$ for $j < N$. Hence $|\sigma| \leq 1$. On $[0, \pi]$, $\sigma \geq 0$: the partial sums $\sum_{j\leq n} \sin((2j+1)t)/(2j+1) = \int_0^t \frac{\sin(2(n+1)u)}{2\sin u}\, du$ are positive on $(0, \pi)$ (Fejér; see [13, §4.1]), and Abel summation with the decreasing coefficients c_j preserves this. So $G_3(t) \geq 0$ for $t \leq \pi$ and $G_3(t) \geq -\pi/(2t) \geq -\frac{1}{2}$ for $t \geq \pi$. The certificate $v_{2j+1} = 2c_j$ satisfies (1) with Ω_3 in place of J_0, so $m_D(\mathbb{R}^3) \leq (1+2N)^{-1}$, and Theorem 2.1(1) gives the first claim. For the second, take $N = \lceil k/2 \rceil$. □

Since at most four points of $\mathbb{R}^3$ have pairwise odd distances, this bound is not a clique bound. The optimal certificates give about $5.3N$.

The proof uses only that consecutive distances have ratio at least q: the arguments satisfy $d_j t \leq Aq^{j-m}$ for $j \leq m$ and $d_j t > Aq^{j-m-1}$ for $j > m$. So the bounds hold for any K positive

algebraic distances $d_0 < \cdots < d_{K-1}$ with $d_{j+1}/d_j \geq q$. Since $(n+1)!/n! = n+1 \geq 32$ for $n \geq 31$, the second parameter set gives a quantitative form of Kahle's question.

Corollary 3.4. *Every coloring of the plane with $k \geq 2$ colors has two points of the same color at distance $n!$ for some $n \leq 31 + \log_{1.75} k$.*

Corollary 3.5. *Let S be a set of positive real algebraic numbers that meets every interval $[X, 2X]$ with $X \geq X_0$; for example the primes, the integers and the powers of 2 ($X_0 = 1$), or the perfect squares ($X_0 = 2$). Then every coloring of the plane with $k \geq 2$ colors has two points of the same color at a distance $s \in S$ with $s \leq 2X_0 k^{7.44}$.*

Proof. Choose $s_0 \in S \cap [X_0, 2X_0]$ and $s_{j+1} \in S \cap [32s_j, 64s_j]$. With $K = \lceil \log_{1.75} k \rceil$, the distances $s_0, \ldots, s_{K-1}$ have consecutive ratios at least 32, so some s_j with $j \leq K-1$ is monochromatic, and $s_{K-1} \leq 2X_0 \cdot 64^{K-1} < 2X_0 k^{6/\log_2 1.75} < 2X_0 k^{7.44}$. For integer-valued polynomials with positive leading coefficient the hypothesis holds for some X_0 depending on the polynomial. □

This makes the theorems of [7, 8] on odd, prime and polynomial distances effective, with polynomial bounds. For odd distances Theorem 3.1 gives the better bound $18k^2$.

For lacunary distances in $\mathbb{R}^3$ the Bessel function is likewise replaced by $\Omega_3(x) = \sin x/x$, which satisfies $\Omega_3 \geq -0.21724$, $\Omega_3(x) \geq 1 - x^2/6$ and $|\Omega_3(x)| \leq 1/x$. The same case analysis with $\lambda = 4$, $v_j = 4\lambda^j$, $A = \frac{1}{2}$ and $\kappa = 2\frac{\lambda/q}{1-\lambda/q} \leq 0.0318$ gives the following, with brackets at least 0.21, -0.04 and $1 - 4(0.21724 + 0.0318) > 0$ in the three cases.

Theorem 3.6. *For real algebraic $q \geq 256$ and $K \geq 1$, $\chi(\mathbb{R}^3, \{1, q, \ldots, q^{K-1}\}) \geq 1 + \frac{4}{3}(4^K - 1)$. Consequently every coloring of $\mathbb{R}^3$ with $k \geq 2$ colors has two points of the same color at a distance $2^i < k^{2.58}$; for this take $q = 8$, $\lambda = 2.24$, $v_j = 1.99\lambda^j$, $A = 1.38$, with brackets $1.4464 - 0.3197 - 0.4867 - 0.6313 > 0$, $1 + 1.99(1 - 0.3197 - 0.4867 - 0.6313) > 0$ and $1 - 1.99(0.21724 + 0.2818) > 0$, so that $\chi(\mathbb{R}^3, \{8^j : j < K\}) \geq 1 + 1.6(2.24^K - 1)$ and $8^{\lceil \log_{2.24} k \rceil - 1} < k^{2.58}$.*

For two distances a direct two-term certificate does better.

Corollary 3.7. *For every real algebraic $d \geq 10^4$, $\chi(\mathbb{R}^2, \{1, d\}) \geq 12$ and $\chi(\mathbb{R}^3, \{1, d\}) \geq 31$. For $d \geq 10^{10}$, $\chi(\mathbb{R}^2, \{1, d\}) \geq 13$.*

Proof. Take $h(t) = 1 + aJ_0(t) + bJ_0(dt)$ and $A > 0$. If $t > A$, then $h \geq 1 - \rho a - b\sqrt{2/(\pi dA)}$. If $t \leq A < dt$, then $h \geq 1 + a(1 - A^2/4) - \rho b$. If $dt \leq A$, both Bessel values are positive. With $\rho = 0.40276$, $A = 0.3$, $a = 2.1976$, $b = 7.8131$ the first two bounds are 0.0011 and 0.0013 for all $d \geq 10^4$, and $1 + a + b > 11$. With $A = 0.05$, $a = 2.4794$, $b = 8.6317$ they are 0.0011 and 0.0013 for $d \geq 10^{10}$, and $1 + a + b > 12$. In $\mathbb{R}^3$, use Ω_3, $\rho = 0.21724$, $1 - A^2/6$ and the envelope $1/(dA)$, with $A = 0.3$, $a = 4.5595$, $b = 25.2703$: the bounds are 0.0010 and 0.0013, and $1 + a + b > 30$. □

Covering a range of d by intervals, each with a single certificate valid for all its values of d, gives the following.

Theorem 3.8. *For every real algebraic $d \in [1.77, 1.92] \cup [7, \infty)$, $\chi(\mathbb{R}^2, \{1, d\}) \geq 7$.*

Proof. For $d \geq 10^4$ this follows from Corollary 3.7. For $d \in [d_1, d_2] \subset [7, 10^4]$ we use $h(t) = 1 + aJ_0(t) + bJ_0(dt)$ with $b \geq 0$ and bound $J_0(dt)$ below by $\min\{J_0(x) : d_1 t \leq x \leq d_2 t\}$, computed from a tabulation of J_0 on $[0, 2050]$ with spacing 10^{-4} and a grid error term, and by $-\sqrt{2/(\pi d_1 t)}$ when $d_1 t > 2000$. Positivity is checked on a t-grid with a Lipschitz margin, and beyond a cutoff T by the envelope bound. Thirteen intervals cover $[7, 10^4]$ and two cover $[1.77, 1.92]$, each with $1 + a + b > 6$. The intervals and coefficients are listed in the accompanying files. All covers in this section were re-verified independently with ball arithmetic (Arb), including the exact decimal endpoints, with margins at least 0.01. □

The same method certifies larger values: $\chi(\mathbb{R}^2, \{1, d\}) \geq 8$ for every algebraic $d \geq 16$, ≥ 9 for $d \geq 40$, and ≥ 10 for $d \geq 150$, using 12, 11 and 10 intervals in $[16, 10^4]$, $[40, 10^4]$ and $[150, 10^4]$ together with Corollary 3.7.

The same certificate shows that measurable sets avoiding $\{1, q, \ldots, q^{K-1}\}$ have density at most $(1 + \frac{22}{15}(2.5^K - 1))^{-1}$. By Bukh's theorem the optimal base for widely separated distances is $1/m_1(\mathbb{R}^2) > 4.04$, but that version gives no explicit distances. Numerically, the optimal certificates in (1) give about $2.9\sqrt{N}$ for odd distances, about 1.3^K for $D = \{1, 2, 4, \ldots, 2^{K-1}\}$, and faster than linear growth for $D = \{1, 4, 9, \ldots, N^2\}$. Rounded and shrunk rational solutions certify the following. Each is checked in floating point on a grid with margin 0.03 against a Lipschitz error of at most 0.005, together with the bound $|J_0(x)| \leq \sqrt{2/(\pi x)}$, valid for all $x > 0$ [12, §13.74], beyond the grid.

Corollary 3.9. *Every coloring of the plane with k colors has two points of the same color at a distance in D, for*

$$\begin{aligned} k = 6: &\quad D = \{1, 3, 5, 7\}; \\ k = 8: &\quad D = \{1, 3, 5, \ldots, 15\}; \\ k = 10: &\quad D = \{2, 3, 5, 7\}; \\ k = 16: &\quad D = \{1, 2, 4, 8, 16, 32\}; \\ k = 18: &\quad D = \{1, 4, 9, \ldots, 64\}. \end{aligned}$$

4. Formal verification

The Lean development builds on the library of [17] and on the formalization of the transfer in [5] (`OAI.HNN.euclidean_transfer_set`).

- `OAI/Soifer` (about 750 lines) proves the weak form of the Furstenberg–Katznelson–Weiss theorem needed here, by Parseval on a torus of side $3T$ and the circle decay bound of [17]: a measurable set in the plane of positive upper density cannot avoid every distance of an unbounded set. Combined with `euclidean_transfer_set` this gives `mono_distance_of_unbounded` (Theorem 2.1(3) for unbounded sets of algebraic distances) and `soifer`, `kahle`:

  ```
  theorem soifer {k : Nat} [NeZero k] (c : EuclideanSpace R (Fin 2) -> Fin k) :
      exists N : Nat, forall n >= N, exists x y, dist x y = (2:R)^n /\ c x = c y
  ```

  and the same with $n!$ in place of 2^n, and for every $n \geq 2$ in place of the plane. The same directory also contains a torus version of the Delsarte bound (`torus_delsarte`) and a lacunary certificate built from $|\hat{\sigma}| \leq 1$, $\operatorname{Re}\hat{\sigma}(\xi) \geq 1 - 2\pi^2|\xi|^2$ and the decay bound of [17], proved by induction on K. Together they give `lacunary_chromatic`: every coloring of the plane avoiding the distances $(10^6)^j$, $j < K$, uses at least $\frac{4}{9}(3/2)^K$ colors. Hence `erdos706_not_polynomial`: for all C and e there is a K-element set of integer distances that no coloring of the plane with at most CK^e colors avoids. By compactness (`erdos706_finite`) the same holds for a finite point set P, as in Erdős's formulation. The certificate works for every $q \geq 10^6$, and with $q = 2^{20}$ it gives `soifer_quantitative`: every coloring of the plane with k colors has two points of the same color at a distance 2^i with $i \leq 20(2\lfloor \log_2 9k \rfloor + 2)$, a formal version, with a weaker exponent, of the last claim of Theorem 3.2. This is a formal negative answer to Erdős Problem #706.

`#print axioms` reports only `propext`, `Classical.choice` and `Quot.sound` for every statement listed. The Fourier certificates of Section 3, other than the lacunary one, and Bukh's theorem used in Theorem 2.1(2) are not formalized.

5. Remarks

Algebraic versus transcendental distances. Bukh [2] wrote that it is very likely that the ordinary chromatic number does not share the exponential growth of the measurable one, and conjectured that $\chi_D(\mathbb{R}^n)$ is bounded when the elements of D are algebraically independent; Davies, McCarty and Pilipczuk [8, Conjecture 5.6] record the stronger form that an algebraically independent unbounded set can be avoided by a finite coloring of the plane. Theorem 2.1 shows that for algebraic distances the exponential growth does transfer. As far as we can see, the only step of the transfer that uses algebraicity is the number-field argument in the proof of [17, Theorem 2.3] (Lemma 3.4 and §3.5 there); the other steps appear to work over any countable real closed subfield $F \subset \mathbb{R}$ with $E = F(i)$. Consequently, if the rigidity theorem held for $E = F(i)$ with F the real closure of $\mathbb{Q}(S)$ for an unbounded algebraically independent set S, then the argument of Theorem 2.1(3) would show that no finite coloring of the plane avoids S, contradicting that conjecture. Either the rigidity theorem is genuinely arithmetic, or the conjecture fails. In dimension one the transfer fails already for $D = \{1, \sqrt{2}\}$: a Hamel-basis parity coloring avoids D with two colors, while no measurable two-coloring does.

References

[1] G. Ambrus, A. Csiszárik, M. Matolcsi, D. Varga, P. Zsámboki, *The density of planar sets avoiding unit distances*, Math. Program. (2023); arXiv:2207.14179.

[2] B. Bukh, *Measurable sets with excluded distances*, Geom. Funct. Anal. 18 (2008), 668–697; arXiv:math/0703856.

[3] E. DeCorte, F. M. de Oliveira Filho, F. Vallentin, *Complete positivity and distance-avoiding sets*, Math. Program. (2022); arXiv:1804.09099.

[4] C. Crocker, *A conditional transfer for colourings of $\mathbb{R}^n$, with the conditional consequence $\chi(\mathbb{R}^3) \geq 7$*, Zenodo (10 October 2026), doi:10.5281/zenodo.23273952.

[5] H. Bass, *A machine-checked proof that three-space is not six-colorable*, preprint (October 2026).

[6] H. T. Croft, K. J. Falconer, R. K. Guy, *Unsolved Problems in Geometry*, Springer, 1991 (corrected reprint 1994).

[7] J. Davies, *Odd distances in colourings of the plane*, Geom. Funct. Anal. (2024); arXiv:2209.15598.

[8] J. Davies, R. McCarty, M. Pilipczuk, *Prime and polynomial distances in colourings of the plane*, arXiv:2308.02483.

[9] J. Davies, *Measurable obstructions for unmeasurable colourings*, arXiv:2610.12301 (2026).

[10] T. F. Bloom, *Erdős Problems*, https://www.erdosproblems.com, Problem #706.

[11] J. Steinhardt, *On coloring the odd-distance graph*, Electron. J. Combin. 16 (2009), N12; arXiv:0908.1452.

[12] G. N. Watson, *A Treatise on the Theory of Bessel Functions*, 2nd ed., Cambridge Univ. Press, 1944.

[13] G. E. Andrews, R. Askey, R. Roy, *Special Functions*, Cambridge Univ. Press, 1999.

[14] P. Erdős, *On the combinatorial problems which I would most like to see solved*, Combinatorica 1 (1981), 25–42.

[15] H. Furstenberg, Y. Katznelson, B. Weiss, *Ergodic theory and configurations in sets of positive density*, in Mathematics of Ramsey Theory, Springer (1990).

[16] E. Naslund, *The chromatic number of $\mathbb{R}^n$ with multiple forbidden distances*, arXiv:2205.12312.

[17] OpenAI, *The Euclidean plane is not five-colorable*, preprint (2026), https://github.com/openai/math, with Lean formalization.

[18] F. M. de Oliveira Filho, F. Vallentin, *Fourier analysis, linear programming, and densities of distance avoiding sets in $\mathbb{R}^n$*, J. Eur. Math. Soc. 12 (2010).